Online Resource 1 for "When can a prediction replace a physical trial? A production-referenced evaluation of design-stage decision rules in deep drawing"

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Abstract

This resource holds the tables behind statements of the paper that its main text summarises: the distances of every rule and per characteristic, the simulation dataset against the measurement and the redundancy of the scans, the fitted Gaussian processes with a sensitivity analysis of the trend GP's priors, the predictors crossed with relation-matched intervals and the coverage of every interval rule, the guard bands available when deciding, the costs, the between-run sensitivity with the trial, and the cross-fitted selection. Every number is written by `scripts/make_tables.py` from a result file of the archived release (Canseven 2026, Zenodo); each table names its files, and `results/DATA_DICTIONARY.md` explains their columns.

1 How to read the tables

The relations and rules are those of the paper (its Table 2 and Section 3.3). A new design is judged with a produced *sibling* (the alternative alone held out), at a new process setting that is *interpolated* (300 kN held out) or *extrapolated* (100 or 500 kN held out), or as a *new family* (one geometry calibrates the rules for the other). Distances are in within-run floors of the characteristic's own geometry, a new family in the floor of the produced family unless stated otherwise; "<1" marks distances below

one floor, and a pair a/b is the decisive and the safe distance. P+RM is the point prediction of rule P with the relation-matched interval; M3n is named M4 in the code. The reference prior of requirement distances weights the 47 distances of the evaluation grid within ±20 floors equally (0, ±0.5 to ±10 in steps of 0.5, ±12, ±15 and ±20), the release-like prior the grid within ±5 floors.

2 Distances

Table S1 gives every rule in every relation, with the interval of its decisive distance and a new family also in its own floor, and Table S2 both distances per characteristic for the main rules and M7. On the cup depth, M1s and M7 falsely accept extrapolated alternatives up to 165 and 118 floors from their limit.

3 Simulation and measurement

Table S3 compares the simulation dataset with the measurement in every cell, at the nominal friction and at the closest grid friction, with the extraction noise of the simulated characteristics; Table S4 gives the redundancy of the scans per series and the components of the floor.

4 Learning and intervals

Table S5 gives the fitted Gaussian processes by relation and Table S6 the trend GP M7n with each prior median halved or doubled. Table S7 crosses eight point predictors with the relation-matched interval and sets them against the rules' own intervals and against M7n calibrated on both families. Tables S8 and S9 give the coverage of every interval rule, overall and per family, characteristic and decision-time stratum.

5 Guard bands

Table S10 gives the guard bands a team can have when it decides, under both priors, with intervals, verdict counts, the worst characteristic and the share correct when a force cell of the family's own production is left out instead; Table S11 gives the share correct per characteristic, and Table S12 the band and trial rate against the width of the prior.

6 Costs, between-run variation and selection

Table S13 gives the cheapest option for every relation, trial cost, prior and ratio of false-reject to false-accept cost; Table S14 the distances and bands with a between-run component added to the truths, and the trial under the same component; Table S15 the rules and bands selected on one geometry and evaluated on the other.

7 Development record

M0–M5 and M3n were defined before the first results; everything else was added afterwards, most of it in response to reviews of earlier versions simulated with a large language model, which are not reviews by a journal. Four simulated rounds led to M0w, Mc, M1n, NN, M2n, M5n, the grouped schemes, the exact estimator of the distances, the noise-variance bound, M1s, M1sn, the jackknife+ intervals, the guard bands with their priors and the new family scored in the produced family's floor. A fifth round, of the version prepared for the journal, led to M7n, M7, Mcd, the relation-matched intervals, the decision-time strata, the per-family and out-of-sample guard bands, the costs, the between-run sensitivity, the cross-fitting, the few-shot transfer and the design-level verdicts, and a sixth, of its revision, to the guard bands available when deciding, the band widened by stated costs, the prior sensitivity of M7n and the trial under between-run variation.

Table S1 All rules: decisive/safe distances (floors) by relation, with the 95 % interval of the decisive distance

Rule	Sibling	Interpolated	Extrapolated	New family	Own floor
M0	108	109	106	104	108
interval	101–115	107–109	101–109	84–112	101–115
M1s†	4.9	19	19	121	122
interval	4.7–6.3	19–20	15–19	119–121	121–123
M2	23/<1	18/2.7	18/9.9	41/26	51/16
interval	22–23	15–18	17–19	39–42	50–52
M5	7.2/<1	19/2.8	19/8.9	35/31	38/28
interval	6.6–8.4	18–20	18–20	31–39	35–39
M7n‡	6.8/<1	7.4/3.4	15/5.9	132/130	132/130
interval	6.0–7.3	7.1–8.1	15–16	131–132	130–132
NN†	3.4	4.7	9.5	130	131
interval	2.0–4.5	4.3–5.1	6.6–11	129–131	129–131
M1sn†	3.5	4.7	8.9	131	130
interval	3.1–4.2	4.3–5.1	8.6–9.5	129–131	129–131
NN+RM‡	6.3/<1	9.6/<1	20/<1	293/4.5	327/4.4
interval	5.6–6.7	9.4–9.8	18–20	293–294	322–331
M1	15	11	25	33	35
interval	12–18	10–11	24–25	31–36	33–37
M1n†	9.1	4.8	12	131	131
interval	5.5–10	4.1–5.3	9.8–13	129–131	129–131
M2n†	19/<1	13/<1	16/9.1	133/128	132/129
interval	15–21	12–13	15–18	132–133	131–133
M5n†	7.2/<1	16/<1	18/8.6	133/128	132/130
interval	6.4–7.5	15–16	16–19	131–133	130–133
M7‡	7.1/<1	22/16	28/17	140/104	144/109
interval	6.1–8.2	22–23	23–54	139–141	144–146
Mc†	27	14	40	43	38
interval	24–29	13–14	39–42	35–48	36–45
Mcd‡	7.2/<1	22/1.6	24/5.8	30/19	28/19
interval	6.1–7.9	20–22	23–31	25–32	28–29
M0w†	119/94	119/94	119/77	117/79	119/94
interval	118–119	118–119	116–120	115–119	118–119
M3	23/<1	29/13	33/17	40/17	49/21
interval	17–26	24–31	25–40	36–45	45–53
M3n	11/<1	15/7.0	13/7.8	135/126	133/128
interval	10–13	13–15	12–14	134–135	131–133
M6†	17/<1	47/<1	56/8.1	35/28	41/25
interval				34–39	
M1sn+RM	7.1/<1	9.6/<1	18/<1	293/4.3	327/4.2
interval	6.6–7.8	9.4–9.8	17–18	293–294	322–331
M1s+RM	11/<1	43/6.9	31/4.8	243/<1	325/<1
interval	11–12	38–43	30–40	243–243	321–331
M5n+RM	6.1/<1	9.6/<1	24/<1	293/4.5	325/4.3
interval	5.2–7.2	9.4–9.8	22–24	293–294	322–331
M5+RM	6.0/<1	21/<1	26/2.2	52/<1	81/<1
interval	5.2–6.8	21–22	26–26	51–53	80–81
M3n+RM	8.1/<1	21/<1	20/<1	292/4.8	325/4.6
interval	7.1–9.9	21–21	17–20	291–293	321–331
M7n+RM	6.5/<1	11/<1	18/<1	293/5.1	327/4.9
interval	5.6–7.4	10–12	17–18	293–294	325–331
M7+RM	6.7/<1	45/6.4	31/<1	248/<1	332/<1
interval	6.0–7.7	39–46	30–40	247–248	327–336

One number for rules that always decide. New family: in the produced family's floor, as step 6 scores it; own floor: in the new family's floor. Intervals over 1000 resamples of the held-out alternatives, fits fixed. M3n is M4 in the code; P+RM: predictor P with the relation-matched interval. †, ‡: added after the first results and in the revisions for this journal. Source: `dec_resolution`, `learn_resolution`, `panel_source_floor`, `learn_source_floor`, `rob_decisions`.

Table S2 Decisive/safe distances (floors) per characteristic

Rule	Mid-side (mm)	Corner (mm)	Wavin. (mm)	Wall (°)	Arm (°)	Depth (mm)	Dome (mm)
Sibling							
M0	38	115	15	28	34	115	32
M1s	2.2	4.3	7.2	3.0	4.2	9.0	5.0
M2	20/<1	6.1/<1	23/<1	9.3/<1	27/<1	28/<1	15/<1
M5	3.4/<1	5.7/1.5	1.9/<1	4.8/1.8	7.2/1.2	5.4/<1	12/5.3
M7n	3.4/<1	3.6/<1	2.5/<1	3.5/<1	6.8/<1	5.7/1.4	12/6.3
M7	3.1/<1	4.6/<1	2.3/<1	3.6/<1	7.6/1.1	5.7/1.5	11/7.0
NN	2.0	4.5	<1	3.4	2.8	3.3	7.0
M1sn	2.6	3.5	3.6	3.0	4.6	3.8	6.0
M3	26/<1	14/<1	21/<1	51/<1	25/<1	26/<1	36/<1
Interpolated							
M0	22	109	15	18	27	115	29
M1s	2.8	1.3	19	<1	31	12	5.7
M2	11/<1	7.1/<1	16/4.2	7.6/2.7	26/<1	14/<1	18/2.8
M5	16/<1	5.1/<1	19/1.3	7.6/3.0	34/<1	17/2.2	17/3.8
M7n	4.5/1.9	6.9/<1	6.1/3.8	3.9/<1	7.9/5.2	8.1/2.7	9.1/5.2
M7	3.6/1.3	5.9/<1	22/16	9.0/<1	49/17	16/10	11/3.6
NN	3.8	1.9	4.8	<1	5.9	4.4	5.3
M1sn	3.8	1.9	4.8	<1	5.9	4.4	5.3
M3	23/15	21/1.8	32/9.0	16/7.1	26/19	19/11	31/6.3
Extrapolated							
M0	38	115	8.6	28	35	109	32
M1s	4.0	2.9	21	2.0	5.9	165	15
M2	18/8.3	5.4/2.2	16/11	6.6/3.2	28/8.4	15/13	18/1.3
M5	20/6.8	5.7/2.0	17/11	7.1/3.2	35/8.4	17/13	16/2.5
M7n	7.9/3.6	12/<1	11/7.5	7.6/<1	19/<1	15/7.7	16/<1
M7	6.6/2.2	9.4/<1	24/18	6.4/<1	15/<1	204/118	54/1.6
NN	6.5	6.7	11	1.9	5.4	15	4.3
M1sn	5.7	2.8	9.3	1.8	9.5	9.9	3.0
M3	34/19	20/11	25/16	23/8.1	44/20	22/13	39/3.0
New family							
M0	40	113	26	29	40	120	33
M1s	15	107	18	125	43	29	23
M2	20/1.9	11/3.4	32/9.8	10/<1	46/34	25/5.9	40/36
M5	5.5/2.1	9.9/3.9	22/21	7.5/3.1	39/32	20/15	40/37
M7n	21/18	170/169	12/11	131/129	48/44	48/44	33/26
M7	17/13	132/77	33/3.7	155/108	50/43	60/26	47/18
NN	20	170	12	130	46	46	28
M1sn	20	170	13	130	47	45	28
M3	17/3.2	15/6.6	32/17	31/5.3	58/40	31/15	40/17
Pooled floor	0.051	0.046	0.0028	0.078	0.11	0.011	0.0070

In the floor of the characteristic's geometry; a new family in the produced family's floor. 18 cases per characteristic with a sibling and for a new family, 6 interpolated, 12 extrapolated. Pooled floor: to read the distances in the unit. Source: `qual_per_characteristic`.

Table S3 The simulation dataset against the measurement per characteristic, geometry and force

Geometry	kN	Measured	$\mu = 0.10$	Floors	Range over μ	Closest μ	Residual	x–y	Rough.
Draw-in, mid-side (mm)									
concave	100	15.14	16.98	37	16.63–17.34	0.15	30.0	0.1	0.1
	300	14.82	15.66	17	14.83–16.67	0.15	0.2		
	500	14.73	14.85	2	13.59–16.14	0.11	−2.4		
convex	100	16.08	18.09	38	17.81–18.41	0.15	33.1	0.2	0.1
	300	15.76	16.91	22	16.17–17.71	0.15	7.7		
	500	15.66	16.05	7	14.78–17.23	0.12	−1.7		
Draw-in, corner (mm)									
concave	100	4.962	8.385	65	8.240–8.525	0.15	62.2		0.1
	300	4.811	7.971	60	7.517–8.328	0.15	51.4		
	500	4.620	7.210	49	5.876–8.046	0.15	23.8		
convex	100	10.00	13.48	116	13.31–13.65	0.15	110.5		0.2
	300	9.876	13.14	109	12.76–13.47	0.15	96.3		
	500	9.668	12.58	97	11.38–13.27	0.15	57.1		
Flange waviness (mm)									
concave	100	0.1028	0.08211	−6	0.05973–0.09609	0.05	−1.8		0.4
	300	0.06655	0.01440	−14	0.01300–0.05128	0.05	−4.2		
	500	0.05241	0.03940	−4	0.01075–0.07766	0.12	0.9		
convex	100	0.07836	0.06604	−6	0.05264–0.06648	0.14	−5.6		4.0
	300	0.06106	0.05964	−1	0.05690–0.08995	0.10	−0.7		
	500	0.06110	0.07239	5	0.05734–0.1112	0.07	1.4		
Wall angle (°)									
concave	100	21.60	20.92	−8	19.77–21.10	0.07	−6.2	2.3	3.1
	300	21.54	20.32	−15	19.44–20.61	0.11	−11.5		
	500	21.56	19.42	−27	19.33–20.92	0.15	−8.0		
convex	100	11.45	10.74	−9	10.01–10.91	0.11	−6.9	1.8	1.7
	300	11.43	10.30	−15	10.30–10.85	0.05	−7.5		
	500	11.50	11.13	−5	10.31–11.20	0.14	−3.9		
Arm angle, cut (°)									
concave	100	0.7594	−0.4379	−10	−1.103 to 0.1517	0.05	−5.2	0.5	0.9
	300	0.4757	−3.702	−36	−4.363 to −1.142	0.05	−13.9		
	500	0.2892	−4.349	−40	−4.451 to −2.928	0.05	−27.7		
convex	100	−4.330	−4.443	−1	−4.687 to −3.104	0.11	0.4	1.1	0.9
	300	−4.947	−5.503	−6	−5.503 to −4.246	0.06	0.0		
	500	−4.720	−4.296	4	−5.586 to −2.301	0.08	0.0		
Cup depth (mm)									
concave	100	28.44	29.35	104	29.35–29.38	0.10	104.3		0.8
	300	28.33	29.34	116	29.31–29.36	0.06	112.1		
	500	28.29	29.24	109	29.00–29.33	0.15	82.3		
convex	100	28.82	29.88	86	29.81–29.88	0.05	80.8		0.7
	300	28.72	29.64	75	29.47–29.80	0.15	61.5		
	500	28.66	29.37	58	29.12–29.64	0.15	37.9		
Bottom dome, cut (mm)									
concave	100	0.02971	0.1336	30	0.01244–0.1336	0.15	0.2		9.8
	300	0.03595	0.1387	30	0.03240–0.1753	0.15	−1.0		
	500	0.04309	0.08286	12	−0.004812 to 0.1545	0.08	2.1		
convex	100	0.1162	0.04951	−8	−0.002123 to 0.06661	0.12	−6.0		2.4
	300	0.1047	0.01768	−11	0.001869–0.08873	0.12	−1.9		
	500	0.09375	0.02753	−8	−0.0008228 to 0.08175	0.11	−1.5		

Values in the unit of the characteristic (signed); simulation at 0.99 mm. $\mu = 0.10$: the nominal simulation; floors: its difference from the measurement. Range over μ: over the friction grid 0.05–0.15; closest μ: grid friction nearest the measurement, residual there (floors). x–y, Rough.: extraction noise of the simulated characteristic, standard deviation of the difference between the two directions of the quarter model and roughness along the friction grid (floors). Levels depend on the assumed blank (210 mm) and the scanner's scales; the mid-side force effect, the arm sign and the dome ranking do not and hold over the whole friction–thickness grid, the concave wall-angle trend only at the nominal friction. Mc's friction fits end at the upper edge of the grid in 72 of 72 within-family calibrations. Source: `sim_validation`, `sim_mc_friction`.

Table S4 Redundancy of the scans per series, and the components of the floor

Series	$W_y - W_x$	σ_y/σ_x	r_{xy}	Diagonals	N−EW	r_{EW}
concave/100/coarse	2.49	5.9	0.01	1.84	0.12	0.09
concave/100/fine	3.47	3.8	0.20	2.06	0.39	0.27
concave/100/medium	2.69	6.1	-0.12	1.94	0.15	0.08
concave/300/coarse	2.82	4.5	-0.04	1.79	0.12	0.16
concave/300/fine	2.48	4.4	-0.14	1.81	0.20	0.05
concave/300/medium	2.70	5.8	0.06	1.75	0.12	0.17
concave/500/coarse	3.29	4.7	0.00	1.74	0.13	0.04
concave/500/fine	2.86	5.1	-0.01	1.64	-0.05	-0.10
concave/500/medium	2.94	3.7	0.01	1.67	0.09	0.01
convex/100/coarse	3.46	17.7	0.05	1.94	-0.92	-0.66
convex/100/fine	2.29	9.5	-0.21	1.96	-0.96	-0.57
convex/100/medium	2.98	16.2	0.00	1.96	-0.92	-0.71
convex/300/coarse	2.33	7.7	0.09	2.17	-1.23	-0.55
convex/300/fine	2.44	7.9	-0.02	1.89	-0.98	-0.53
convex/300/medium	2.43	5.3	0.01	1.87	-0.92	-0.54
convex/500/coarse	2.32	5.4	0.02	2.16	-1.23	-0.60
convex/500/fine	2.40	6.1	-0.00	1.96	-1.64	-0.62
convex/500/medium	2.64	8.1	0.02	2.13	-1.33	-0.49

Components of the floor (fractions of the floor), concave/convex

Characteristic	σ_{b}/F	σ_{w}/F	Drift ratio
Draw-in, mid-side	0.32/0.34	0.91/0.66	3.3
Draw-in, corner	0.31/0.35	0.65/0.77	4.5
Flange waviness	0.33/0.30	0.55/1.31	2.5
Wall angle	0.32/0.26	0.68/1.39	2.1
Arm angle, cut	0.36/0.31	0.46/1.22	2.6
Cup depth	0.31/0.34	0.65/0.52	3.9
Bottom dome, cut	0.33/0.36	0.76/0.55	3.9

$W_y - W_x$: flange width in the direction of motion minus that along the laser line (mm); σ_y/σ_x: ratio of their part-to-part standard deviations; r_{xy}: their correlation; diagonals: difference of the two flange diagonals (mm); N−EW: north wall angle minus the mean of east and west (°); r_{EW}: correlation of east and west. σ_{b}: standard deviation between batch means; σ_{w}: within batches; drift ratio: floor over that of the series in random order. Source: `meas_scan`, `panel_floor_protocol`, `sens_drift`.

Table S5 Fitted Gaussian processes by relation: hyperparameters and where their centres lie

Relation	Rule	Fits	ℓ_F	At bound	Noise	Δmean	≤0.01	ΔNN	ΔM1sn	HW
Sibling	M5	126	0.13	45	1.00	2.6	0	0.27	0.92	2.2
	M5n	126	1.76	22	1.00	2.1	20	0.28	0.88	2.3
	M7n	126	1.43	–	3.6	2.6	0	0.40	1.0	1.6
	M7	126	1.98	–	3.6	2.6	0	0.48	1.0	1.6
Interpolated	M5	14	0.10	64	1.00	4.1	0	4.1	4.1	6.4
	M5n	14	0.10	79	1.00	0	95	0	0	5.8
	M7n	14	1.92	–	4.0	0.36	2	0.36	0.36	2.0
	M7	14	1.92	–	4.0	1.4	0	1.4	1.4	3.0
Extrapolated	M5	28	0.10	54	1.00	6.8	0	6.1	4.9	4.6
	M5n	28	0.10	68	1.00	0	93	1.3	3.9	3.7
	M7n	28	1.95	–	3.1	4.2	0	2.9	0.33	2.8
	M7	28	1.95	–	3.1	3.6	1	3.1	2.1	3.2
New family	M5	14	0.21	43	1.00	24	0	21	22	1.9
	M5n	14	1.77	21	1.00	1.9	13	0.22	0.66	2.0
	M7n	14	1.51	–	3.7	2.2	1	0.37	0.83	1.2
	M7	14	1.87	–	4.0	6.8	1	5.0	4.8	7.3
Pooled interp.	M5	14	2.12	14	1.00	20	0	5.3	20	4.9
	M5n	14	9.33	0	1.00	22	0	0.81	22	3.0
	M7n	14	1.52	–	2.9	19	0	1.1	19	2.0
	M7	14	1.91	–	3.0	19	0	1.3	19	2.2
Pooled extrap.	M5	28	1.92	11	1.00	18	0	5.3	25	6.3
	M5n	28	9.07	0	1.00	17	4	1.7	25	4.4
	M7n	28	1.25	–	2.9	19	0	3.3	29	3.2
	M7	28	1.46	–	2.8	24	0	2.6	30	4.3

ℓ_F: median force length scale (100 kN). At bound: fits with ℓ_F at its lower bound (%). Noise: fitted noise over its bound (M5, M5n) or nugget over the sampling error of the truths (M7n, M7), median. Δmean, ΔNN, ΔM1sn: median distance (floors) of the centre from the calibration mean, NN and M1sn; ≤0.01: cases within 0.01 floor of the calibration mean (%). HW: median half-width (floors). Pooled: both families calibrate, with a family term. Source: `learn_gp_diagnostics`.

Table S6 The trend GP M7n with each of its four prior medians halved or doubled

	Sibling			Interpolated				Extrapolated			
Prior median	δ_d/δ_s	Cov	HW	δ_d/δ_s	Cov	HW	ΔM1sn	δ_d/δ_s	Cov	HW	ΔM1sn
reference	6.8/<1	85	1.6	7.4/3.4	52	2.0	0.36	15/5.9	56	2.8	0.33
GP standard deviation x0.5	6.9/<1	84	1.6	7.2/3.5	50	1.8	0.31	15/6.2	50	2.3	0.34
GP standard deviation x2	7.1/<1	84	1.6	7.8/3.3	57	2.2	0.38	16/5.5	60	3.2	0.32
force length scale x0.5	7.1/<1	84	1.6	8.6/2.8	60	2.8	0.25	16/5.5	62	3.8	0.41
force length scale x2	7.1/<1	81	1.6	7.2/3.7	50	1.4	0.38	15/6.4	48	2.1	0.33
lubrication length scale x0.5	7.0/<1	83	1.7	7.3/3.6	50	1.7	0.37	15/6.3	48	2.2	0.30
lubrication length scale x2	6.9/<1	83	1.5	8.2/3.2	57	2.4	0.23	17/5.2	61	3.5	0.37
nugget standard deviation x0.5	6.7/<1	80	1.4	7.3/3.5	50	1.9	0.36	15/5.8	55	2.8	0.38
nugget standard deviation x2	7.7/<1	86	1.8	7.6/3.3	55	2.1	0.30	15/5.9	54	3.0	0.33

Reference medians: force length scale 200 kN, lubrication length scale one pattern step, GP standard deviation two floors, nugget standard deviation one floor; one median changed at a time, every calibration set refitted. Cov: coverage of the 90 % interval (%); HW: median half-width (floors); ΔM1sn: median distance of the centre from the force trend (floors). For a new family every variant decides beyond 131 floors and covers no truth. The reference row reproduces the stored intervals exactly. Source: `learn_m7n_priors`.

Table S7 Point predictors crossed with the relation-matched interval, against the rules' own intervals and a trend GP calibrated on both families

Predictor and interval	δ_d/δ_s	FA	Cov	IS	q (c/v)	n
Sibling						
NN+RM	6.3/<1	<1	84	11	4.4/2.0	63
M1sn+RM	7.1/<1	<1	88	9	3.9/3.3	63
M1s+RM	11/<1	<1	90	15	4.8/6.3	63
M5n+RM	6.1/<1	<1	88	9	3.6/2.6	63
M5+RM	6.0/<1	<1	86	11	3.8/2.1	63
M3n+RM	8.1/<1	<1	88	13	5.8/2.7	63
M7n+RM	6.5/<1	<1	91	11	4.4/2.4	63
M7+RM	6.7/<1	<1	89	11	4.4/2.3	63
M2 (own)	23/<1	<1	89	19		
M3 (own)	23/<1	<1	96	21		
M5 (own)	7.2/<1	<1	86	10		
M7n (own)	6.8/<1	<1	85	10		
M7n, both families	8.5/<1	<1	88	8		
Interpolated						
NN+RM	9.6/<1	<1	88	12	5.3/4.2	21
M1sn+RM	9.6/<1	<1	88	12	5.3/4.2	21
M1s+RM	43/6.9	18	86	76	30.4/12.2	21
M5n+RM	9.6/<1	<1	88	12	5.3/4.2	21
M5+RM	21/<1	<1	93	24	9.7/11.6	21
M3n+RM	21/<1	<1	86	26	9.6/11.4	21
M7n+RM	11/<1	<1	90	14	6.5/4.7	21
M7+RM	45/6.4	20	86	80	32.5/12.4	21
M2 (own)	18/2.7	3.7	71	27		
M3 (own)	29/13	13	43	98		
M5 (own)	19/2.8	2.4	71	29		
M7n (own)	7.4/3.4	2.2	52	26		
M7n, both families	11/1.3	2.7	86	16		
Extrapolated						
NN+RM	20/<1	1.5	93	24	9.1/11.0	42
M1sn+RM	18/<1	<1	88	19	9.2/8.6	42
M1s+RM	31/4.8	155	86	155	9.6/21.0	42
M5n+RM	24/<1	2.1	89	29	10.8/12.8	42
M5+RM	26/2.2	2.4	85	40	11.5/14.5	42
M3n+RM	20/<1	1.5	93	24	8.9/10.6	42
M7n+RM	18/<1	<1	89	19	9.2/8.3	42
M7+RM	31/<1	151	89	149	9.6/21.1	42
M2 (own)	18/9.9	8.7	44	79		
M3 (own)	33/17	13	43	120		
M5 (own)	19/8.9	8.4	46	75		
M7n (own)	15/5.9	7.5	56	41		
M7n, both families	15/3.1	3.9	79	21		
New family						
NN+RM	293/4.5	4.1	86	362	124.4/168.3	63
M1sn+RM	293/4.3	3.9	86	362	124.5/168.1	63
M1s+RM	243/<1	5.9	93	252	118.5/124.3	63
M5n+RM	293/4.5	3.9	86	362	124.5/168.4	63
M5+RM	52/<1	<1	91	78	33.5/34.9	63
M3n+RM	292/4.8	4.5	86	361	123.9/167.5	63
M7n+RM	293/5.1	4.0	86	363	124.4/168.5	63
M7+RM	248/<1	3.6	93	253	121.9/125.7	63
M2 (own)	41/26	27	47	124		
M3 (own)	40/17	18	40	134		
M5 (own)	35/31	33	12	188		
M7n (own)	132/130	129	0	1043		

P+RM: the prediction of P (for interval rules the interval centre) with the 90 % split-conformal quantile q of P's absolute errors (floors) over the n cases of the same relation in the other family; c/v: the quantile borrowed by the concave and by the convex cup. These n cases come from one family and from one or a few shared fits, so the interval carries no coverage guarantee. Own: the rule's own interval. Both families: M7n calibrated on both families with a family term. FA: safe distance on the false-accept side; Cov: coverage (%); IS: mean interval score at $\alpha = 0.1$ (floors). A new family in the produced family's floor. Source: `learn_resolution`, `qual_coverage`.

Table S8 Interval rules: coverage and interval score by relation

		Sibling		Interpolated		Extrapolated		New family	
Rule	Level	Cov	IS	Cov	IS	Cov	IS	Cov	IS
M2	0.78	89 (83–94)	19	71	27	44	79	47	124
M2n	0.78	90 (82–95)	15	93	14	33	59	12	969
M3	0.78	96 (93–98)	21	43	98	43	120	40	134
M3n	0.78	89 (81–96)	11	36	65	48	48	7	989
M5	0.90	86 (77–94)	10	71	29	46	75	12	188
M5n	0.90	85 (76–93)	9	93	16	36	55	0	1024
M7	0.90	83 (73–93)	10	57	81	58	147	17	698
M7n	0.90	85 (75–94)	10	52	26	56	41	0	1043
Mcd	0.90	83 (72–92)	10	79	24	67	50	11	139
NN+RM	0.90	84 (74–94)	11	88	12	93	24	86	362
M1sn+RM	0.90	88 (81–94)	9	88	12	88	19	86	362
M7n+RM	0.90	91 (86–96)	11	90	14	89	19	86	363
M6	0.90	93 (87–98)	15	93	38	61	81	32	145
NN	point	–	24	–	42	–	81	–	1067
M1sn	point	–	32	–	42	–	77	–	1067
M1s	point	–	43	–	122	–	217	–	860

Level: with a sibling, guaranteed for the largest residual and jackknife+ ($n = 8$); for a new setting or family their levels are nominal, a held-out force or family not being exchangeable with the calibration alternatives; model-based intervals and P+RM are nominal (with a single calibration family, P+RM has no guarantee even if the families were exchangeable). Cov: coverage (%), with a sibling with a 95 % interval over 2000 resamples of the alternatives within each geometry; for a new setting or family the alternatives of one held-out cell or family share one fit, so no interval over alternatives is given. IS: mean interval score (Gneiting and Raftery 2007) at $\alpha = 0.1$, floors; lower is better; for point rules $20\,|e|$. M6: conformal margin on standardised residuals of the calibration set. Source: `qual_coverage`, `qual_m6_coverage`.

Table S9 Coverage (%) per family, characteristic and decision-time stratum

Relation	Rule	Conc.	Conv.	Mid-side	Corner	Wavin.	Wall	Arm	Depth	Dome	Not res.	Res.
Sibling	M5	90	81	89	89	89	83	83	83	83	89	80
	M5n	86	84	89	83	89	72	89	89	83	88	80
	M7n	83	87	83	89	100	89	72	83	78	89	78
	M3	95	97	100	100	100	89	94	94	94	96	96
	Mcd	83	83	78	83	100	94	67	78	78	87	76
	NN+RM	95	73	100	89	100	89	72	78	61	95	69
	M7n+RM	95	87	100	94	100	100	89	94	61	96	84
Interpolated	M5	71	71	100	100	50	50	100	67	33		
	M5n	100	86	100	100	100	100	67	100	83		
	M7n	57	48	17	100	0	100	67	33	50		
	M3	33	52	0	67	67	67	50	17	33		
	Mcd	100	57	100	100	50	50	100	83	67		
	NN+RM	100	76	100	100	67	100	67	100	83		
	M7n+RM	100	81	100	100	83	100	67	100	83		
Extrapolated	M5	45	48	0	50	50	50	58	50	67		
	M5n	26	45	33	0	33	67	25	17	75		
	M7n	50	62	33	92	0	67	92	17	92		
	M3	48	38	17	17	50	42	67	25	83		
	Mcd	64	69	25	75	67	100	25	75	100		
	NN+RM	86	100	100	100	75	100	100	75	100		
	M7n+RM	95	83	100	100	50	100	92	83	100		

Conc., Conv.: held-out alternatives of the concave and the convex cup (63 cases with a sibling, 21 interpolated, 42 extrapolated each). 18 cases per characteristic with a sibling, 6 interpolated, 12 extrapolated. Not res., Res.: the produced siblings do not or do differ resolvably from each other (75 and 51 cases). Source: `qual_coverage`.

Table S10 Guard bands available when deciding, by relation and prior, validated by the matching scheme

Relation	Rule	g (c/v)	Correct	c/v	T	Decided	FA	Worst	Cell out
Reference prior									
Sibling	NN	0/0.5	95 (94–97)	96/95	2	2808/2841	1.8	Dome 89	
	M1sn	0.5/1.0	95 (94–97)	95/95	7	2675/2687	1.8	Dome 92	
	M2	0	99 (98–100)	100/99	54	1176/1442	0.1	Dome 98	
	M5	0	99 (98–100)	99/98	23	2233/2199	0.3	Dome 94	
	M5n	0	99 (98–100)	99/99	22	2278/2207	0.3	Dome 95	
	M7n	0	98 (97–100)	98/98	19	2318/2326	0.4	Dome 92	
	Mcd	0	98 (97–99)	98/98	19	2303/2366	0.4	Dome 92	
	NN+RM	0	99 (97–99)	100/98	28	1719/2441	0.4	Dome 94	
Interpolated	NN	2.5/1.5	95	97/93	18	725/852	1.0	Wavin. 88	95
	M1sn	2.5/1.5	95	97/93	18	725/852	1.0	Wavin. 88	95
	M2	0.5/0.75	95	95/96	52	439/475	1.4	Dome 87	95
	M5	0/1.75	95	91/99	57	421/405	0.7	Dome 82	95
	M5n	0	99	100/98	53	425/466	0.4	Arm 96	99
	M7n	1.75/0	95	99/92	28	587/791	1.2	Wavin. 89	95
	Mcd	1.25/0	95	100/92	57	283/535	0.1	Dome 87	
	NN+RM	0	99	100/98	42	482/621	0.3	Arm 96	99
Extrapolated	NN	5.75/9.5	95	92/100	58	1002/598	2.0	Depth 85	93
	M1sn	6.5/5.5	95	97/93	47	898/1119	1.7	Wavin. 87	95
	M2	6.25/none	84	84/–	81	711/0	2.2	Arm 62	83
	M5	5.0/none	83	83/–	80	761/0	2.4	Arm 62	81
	M5n	5.0/8.5	95	92/100	68	797/421	1.7	Depth 84	92
	M7n	4.75/2.75	96	98/93	60	769/778	1.7	Wavin. 85	95
	Mcd	4.5/8.0	94	92/99	78	558/296	0.3	Arm 79	
	NN+RM	0/0.5	97	95/100	73	608/421	0.8	Depth 90	97
New family	Mcd	none/18.25	73	–/73	91	0/502	2.4	Dome 60	
	NN+RM	0/none	59	59/–	87	716/0	1.4	Corner 49	
Release-like prior									
Sibling	NN	1.0/3.0	96 (92–98)	95/96	41	1076/487	0.8	Dome 85	
	M1sn	2.25/3.0	95 (93–97)	95/95	47	785/610	0.9	Dome 89	
	M2	0	97 (94–99)	99/95	76	337/300	0.3	Dome 94	
	M5	0/6.0	95 (91–100)	97/85	67	771/99	0.4	Dome 69	
	M5n	0	96 (93–99)	96/96	44	806/662	0.6	Dome 82	
	M7n	5.5/7.25	84 (82–100)	85/82	91	119/113	0.4	Dome 54	
	Mcd	5.5/5.0	84 (82–100)	85/83	91	123/127	0.4	Dome 59	
	NN+RM	0/1.0	96 (92–100)	97/96	72	269/458	0.3	Dome 85	
Interpolated	NN	4.75/3.5	93	100/90	67	93/196	0.8	Arm 83	93
	M1sn	4.75/3.5	93	100/90	67	93/196	0.8	Arm 83	93
	M2	3.25/2.25	94	100/89	85	59/76	0.8	Wavin. 85	94
	M5	1.5/3.25	88	81/100	84	96/48	0.6	Dome 75	88
	M5n	1.5/0	93	100/91	84	31/111	0.8	Arm 82	93
	M7n	4.25/1.75	87	100/83	74	46/179	1.2	Arm 73	87
	Mcd	4.75/0	77	–/77	80	0/179	0.2	Wall 67	
	NN+RM	0.5/0	95	100/93	77	51/154	0.6	Arm 84	95
Extrapolated	NN	8.5/14.5	75	75/100	92	138/1	2.0	Depth 64	72
	M1sn	8.5/8.25	94	96/92	84	132/154	0.7	Arm 92	96
	M2	10.25/none	72	72/–	91	158/0	2.3	Arm 52	62
	M5	9.25/none	67	67/–	91	165/0	2.6	Arm 52	61
	M5n	8.0/12.5	77	75/100	93	110/11	1.6	Depth 61	77
	M7n	7.0/6.25	93	97/89	90	90/80	0.7	Depth 91	94
	Mcd	8.0/10.5	83	78/100	95	69/21	0.0	Arm 67	
	NN+RM	0/5.25	75	75/–	93	121/0	1.7	Depth 64	73
New family	Mcd	none/19	60	–/60	93	0/176	2.7	Dome 56	
	NN+RM	0/none	52	52/–	86	362/0	2.8	Corner 48	

g: band (floors, concave/convex) from the family's own sibling cases or, for a new setting or family, from the same relation in the other family; none: no band up to 20 floors qualifies. Correct: share correct among decided verdicts (%), pooled over the families; with a sibling one alternative left out within its family (95 % cluster interval), otherwise the other family's band applied, without an interval (the alternatives of a held-out cell share one fit). c/v: per family. T: share sent to trial (%). Decided: per family, of reference prior 2808/2939, 935/978, 1873/1961, 2737/2955; release-like prior 1310/1323, 439/441, 871/882, 1303/1323 verdicts (sibling, interpolated, extrapolated, new family). FA: false accepts (% of verdicts). Worst: the characteristic with the lowest share correct (%). Cell out: share correct with the band from both families' cases except the held-out geometry–force cell (interpolating, the other family's cell: the band of Correct). New family: rows where a rule decides. Source: `qual_guard_decision`, `qual_guard_oos`.

Table S11 Share correct (%) per characteristic under the guard bands available when deciding

Prior, relation	Rule	Mid-side	Corner	Wavin.	Wall	Arm	Depth	Dome
Reference, Sibling	NN	97	96	99	97	94	95	89
	M1sn	97	98	94	98	95	93	92
	M2	100	99	99	99	100	99	98
	M5	100	99	100	99	99	98	94
	M7n	99	99	100	100	98	98	92
	NN+RM	100	98	100	99	98	99	94
Interpolated	NN	97	100	88	100	92	92	94
	M1sn	97	100	88	100	92	92	94
	M2	100	100	90	94	100	99	87
	M5	100	100	100	92	100	98	82
	M7n	97	100	89	100	90	98	89
	NN+RM	100	100	98	100	96	100	98
Extrapolated	NN	99	100	90	100	100	85	100
	M1sn	100	100	87	100	88	93	100
	M2	93	100	78	100	62	73	100
	M5	94	100	72	100	62	70	100
	M7n	100	100	85	100	100	96	100
	NN+RM	100	100	96	100	100	90	100
Release-like, Sibling	NN	97	99	100	99	95	95	85
	M1sn	98	97	95	99	94	92	89
	M2	99	98	98	94	99	96	94
	M5	100	100	100	99	95	90	69
	M7n	100	100	100	100	73	76	54
	NN+RM	100	100	100	100	93	99	85
Interpolated	NN	100	100	87	100	83	100	90
	M1sn	100	100	87	100	83	100	90
	M2	–	–	85	100	–	–	97
	M5	–	–	–	87	–	–	75
	M7n	98	–	81	–	73	100	81
	NN+RM	100	–	94	–	84	100	89
Extrapolated	NN	–	–	77	–	–	64	–
	M1sn	100	–	94	–	92	94	–
	M2	100	–	88	–	52	74	–
	M5	–	–	77	–	52	68	–
	M7n	–	–	92	–	–	91	–
	NN+RM	–	–	81	–	–	64	–

Bands and validation as in Table S10; –: fewer than 20 decided verdicts. Source: `qual_guard_decision`.

Table S12 Guard band (floors) and trial rate (%, in brackets) against the half-width of the prior

Relation	Rule	±3	±5	±10	±20	±50
Sibling	NN	8.0 (99)	2.0 (37)	0.5 (5)	0.25 (2)	0 (0)
	M1sn	4.5 (89)	2.75 (50)	1.0 (10)	0.75 (7)	0.5 (4)
	M2	0.5 (85)	0 (75)	0 (60)	0 (54)	0 (48)
	M5	6.5 (99)	0 (45)	0 (26)	0 (23)	0 (20)
	M7n	7.25 (99)	6.75 (99)	0 (22)	0 (19)	0 (17)
	NN+RM	6.0 (100)	0.5 (65)	0 (31)	0 (28)	0 (24)
Interpolated	NN	5.0 (86)	4.25 (68)	2.5 (25)	2.0 (18)	1.5 (11)
	M1sn	5.0 (86)	4.25 (68)	2.5 (25)	2.0 (18)	1.5 (11)
	M2	3.25 (89)	2.75 (84)	1.25 (62)	0.75 (53)	0 (42)
	M5	3.5 (91)	2.75 (86)	1.0 (62)	0.5 (54)	0 (45)
	M7n	4.75 (93)	3.75 (82)	1.25 (36)	0.75 (27)	0.25 (20)
	NN+RM	0.75 (88)	0 (75)	0 (48)	0 (42)	0 (37)
Extrapolated	NN	14.75 (99)	14.25 (98)	8.25 (69)	7.0 (55)	5.75 (40)
	M1sn	9.0 (90)	8.5 (84)	6.75 (59)	6.0 (48)	5.25 (37)
	M2	none (100)	none (100)	none (100)	none (100)	17.5 (83)
	M5	none (100)	none (100)	none (100)	none (100)	17.75 (83)
	M7n	7.25 (94)	6.75 (91)	4.25 (68)	3.25 (56)	2.25 (44)
	NN+RM	5.5 (99)	5.25 (98)	0 (79)	0 (72)	0 (63)

Prior: the evaluation grid of requirement distances within the half-width, equal weights; ±20 is the reference prior and ±5 the release-like one. Bands pooled over the families, in sample, to show the dependence on the prior. No rule qualifies for a new family. Source: `qual_guard_prior_width`.

Table S13 Cheapest option and its expected cost per verdict by relation, trial cost, prior and cost ratio

Relation	c_T	$c_{FR} = 0.25$	$c_{FR} = 0.5$	$c_{FR} = 1$
Reference prior				
Sibling	0.02	M5n 0.008	M1sn 0.009	M1sn 0.009
	0.05	M5n 0.015	M5n 0.017	M5n 0.020
	0.1	NN 0.022	NN 0.026	NN 0.031
	0.2	NN 0.027	NN 0.033	NN 0.042
	0.5	NN 0.034	NN 0.041	NN 0.054
Interpolated	0.02	NN* 0.009	NN* 0.010	NN* 0.010
	0.05	NN* 0.022	NN* 0.022	NN* 0.023
	0.1	NN* 0.033	NN* 0.039	NN* 0.043
	0.2	NN* 0.051	NN* 0.059	NN* 0.074
	0.5	NN* 0.104	NN* 0.111	NN* 0.127
Extrapolated	0.02	M1sn 0.014	M1sn 0.014	M1sn 0.014
	0.05	M1sn 0.034	M1sn 0.034	M1sn 0.035
	0.1	M1sn 0.066	M1sn 0.067	M1sn 0.067
	0.2	M1sn 0.115	M1sn 0.117	M1sn 0.120
	0.5	M1sn 0.258	M1sn 0.259	M1sn 0.263
New family	0.02	trial 0.020	trial 0.020	trial 0.020
	0.05	trial 0.050	trial 0.050	trial 0.050
	0.1	trial 0.100	trial 0.100	trial 0.100
	0.2	trial 0.200	trial 0.200	trial 0.200
	0.5	trial 0.500	trial 0.500	trial 0.500
Release-like prior				
Sibling	0.02	M5n 0.017	M2 0.017	M2 0.018
	0.05	M5n 0.032	M5n 0.035	M5n 0.041
	0.1	NN 0.052	NN 0.056	NN 0.065
	0.2	NN 0.089	NN 0.093	NN 0.102
	0.5	NN 0.200	NN 0.204	NN 0.213
Interpolated	0.02	NN* 0.017	NN* 0.017	NN* 0.017
	0.05	NN* 0.040	NN* 0.041	NN* 0.041
	0.1	NN* 0.076	NN* 0.078	NN* 0.080
	0.2	NN* 0.144	NN* 0.146	NN* 0.150
	0.5	NN* 0.350	NN* 0.352	NN* 0.356
Extrapolated	0.02	M1sn 0.018	M1sn 0.018	M1sn 0.018
	0.05	M1sn 0.045	M1sn 0.045	M1sn 0.045
	0.1	M1sn 0.089	M1sn 0.089	M1sn 0.089
	0.2	M1sn 0.174	M1sn 0.174	M1sn 0.175
	0.5	M1sn 0.427	M1sn 0.428	M1sn 0.429
New family	0.02	trial 0.020	trial 0.020	trial 0.020
	0.05	trial 0.050	trial 0.050	trial 0.050
	0.1	trial 0.100	trial 0.100	trial 0.100
	0.2	trial 0.200	trial 0.200	trial 0.200
	0.5	trial 0.500	trial 0.500	trial 0.500

Costs relative to a false accept. Each rule with its cost-minimising band under step 5's rule (from its 95 % band up to 20 floors; without a 95 % band, every design to trial), chosen in sample and pooled over the families; trial: sending every design to trial (cost c_T) is cheapest; NN*: NN and M1sn, which coincide at 300 kN. Differences of a few thousandths are not resolved by this design. Trials are priced as error-free: the errors of a 20-part trial cost 0.017 per verdict under the reference prior and 0.037 under the release-like prior ($c_{FR} = 0.5$); both priors place 47 % of the verdicts on failing designs. Source: `qual_costs`, `qual_cost_cheapest`, `qual_trial_cost`.

Table S14 Decisive distance and guard band (floors, in brackets) with a between-run component added to the truths, and the trial under the same component

Relation	Rule	$\tau = 0$	1	2	3	bound
Sibling	NN	3.4 (0.25)	3.5 (0.5)	4.5 (1.5)	5.8 (2.9)	4.1 (1.0)
	M1sn	3.5 (0.75)	3.9 (1.0)	4.8 (2.0)	6.1 (3.25)	4.4 (1.5)
	M1s	4.9 (2.0)	5.6 (2.25)	6.2 (3.0)	7.1 (4.25)	6.1 (3.0)
	M2	23 (0)	23 (0)	23 (0)	23 (0.25)	23 (0)
	M5	7.2 (0)	7.5 (0)	8.3 (0)	9.3 (0.75)	8.3 (0)
	M7n	6.8 (0)	7.0 (0)	7.7 (0)	8.8 (1.0)	7.9 (0)
	record rule best	100	100	100	100	100
Interpolated	NN	4.7 (2.0)	5.0 (2.25)	5.5 (3.0)	6.6 (4.0)	5.1 (2.75)
	M1sn	4.7 (2.0)	5.0 (2.25)	5.5 (3.0)	6.6 (4.0)	5.1 (2.75)
	M1s	19 (none)	20 (none)	21 (none)	21 (none)	19 (none)
	M2	18 (0.75)	18 (0.75)	19 (1.4)	20 (2.5)	19 (1.25)
	M5	19 (0.5)	20 (0.75)	21 (1.25)	22 (2.25)	20 (0.9)
	M7n	7.4 (0.75)	7.6 (1.0)	8.4 (1.5)	9.5 (2.6)	8.4 (1.25)
	record rule best	100	100	100	100	100
Extrapolated	NN	9.5 (7.0)	9.5 (7.25)	9.5 (7.9)	10 (8.9)	9.7 (7.5)
	M1sn	8.9 (6.0)	8.8 (6.25)	9.2 (6.75)	9.9 (8.0)	9.0 (6.5)
	M1s	19 (none)	19 (none)	19 (none)	18 (none)	19 (none)
	M2	18 (none)	18 (none)	18 (none)	19 (none)	18 (none)
	M5	19 (none)	19 (none)	19 (none)	20 (none)	19 (none)
	M7n	15 (3.25)	15 (3.75)	16 (4.25)	16 (5.25)	16 (3.75)
	record rule best	100	100	100	100	100
New family	NN	130 (none)	131 (none)	132 (none)	133 (none)	131 (none)
	M1sn	131 (none)	131 (none)	132 (none)	133 (none)	131 (none)
	M1s	121 (none)	121 (none)	121 (none)	121 (none)	121 (none)
	M2	41 (none)	41 (none)	41 (none)	41 (none)	41 (none)
	M5	35 (none)	35 (none)	35 (none)	35 (none)	35 (none)
	M7n	132 (none)	132 (none)	133 (none)	134 (none)	132 (none)
	record rule best	0	0	0	0	0
The trial, same draws (any relation)						
	first 20 parts	1.4	2.1	3.6	5.1	3.0
	first 50 parts	0.90	1.9	3.5	5.1	2.9

Medians over 200 draws (bands over 50, pooled over the families); τ: standard deviation of the component (floors); bound: each characteristic's variation between series, $\tau = 1.2$–$9.8/2.77$. Record rule best: share of draws in which NN, M1sn or M1n is the most decisive rule (%). The trial saw the observed run; a future run's 95th percentile differs from it by the same component. Source: `qual_between_run`, `qual_between_run_trial`.

Table S15 Rules and guard bands selected on one geometry and evaluated on the other

Relation	Criterion	Selected on	Rule	In sample	Other	g	Correct	T
Sibling	decisive, all rules	concave	NN	2.0	3.6			
Sibling	decisive, all rules	convex	NN	3.6	2.0			
Sibling	decisive, pre-specified	concave	M5	6.5	8.5			
Sibling	decisive, pre-specified	convex	M5	8.5	6.5			
Sibling	safe, all rules	concave	M1sn+RM	0	0.40			
Sibling	safe, all rules	convex	M1s+RM	0	1.1			
Sibling	safe, pre-specified	concave	M3	0	0			
Sibling	safe, pre-specified	convex	M3	0	0			
Sibling	step 6	concave	NN	2.0	3.6	0	94	0
Sibling	step 6	convex	NN	4.1	2.5	0.5	97	4
Interpolated	decisive, all rules	concave	M1n	3.6	5.8			
Interpolated	decisive, all rules	convex	M1sn	5.3	4.2			
Interpolated	decisive, pre-specified	concave	M1	12	6.8			
Interpolated	decisive, pre-specified	convex	M1	6.8	12			
Interpolated	safe, all rules	concave	M1s+RM	0	18			
Interpolated	safe, all rules	convex	M3n+RM	0	1.8			
Interpolated	safe, pre-specified	concave	M2	2.6	3.7			
Interpolated	safe, pre-specified	convex	M5	1.9	3.1			
Interpolated	step 6	concave	M1n	5.1	7.3	1.5	93	13
Interpolated	step 6	convex	M1sn	7.8	6.8	2.5	97	22
Extrapolated	decisive, all rules	concave	M1sn	8.7	8.9			
Extrapolated	decisive, all rules	convex	NN	8.7	11			
Extrapolated	decisive, pre-specified	concave	M3n	12	13			
Extrapolated	decisive, pre-specified	convex	M3n	13	12			
Extrapolated	safe, all rules	concave	M1sn+RM	0	0.25			
Extrapolated	safe, all rules	convex	M1s+RM	0	11			
Extrapolated	safe, pre-specified	concave	M3n	9.9	5.1			
Extrapolated	safe, pre-specified	convex	M3n	5.1	9.9			
Extrapolated	step 6	concave	M1sn	14	14	5.5	93	43
Extrapolated	step 6	convex	NN	14	17	5.75	92	47
New family	decisive, all rules	concave	Mcd	29	33			
New family	decisive, all rules	convex	M1	33	32			
New family	decisive, pre-specified	concave	M1	32	33			
New family	decisive, pre-specified	convex	M1	33	32			
New family	safe, all rules	concave	M5+RM	0.05	0			
New family	safe, all rules	convex	M1s+RM	0	5.9			
New family	safe, pre-specified	concave	M3	18	17			
New family	safe, pre-specified	convex	M3	17	18			
New family	step 6	concave	Mcd	43	38	18.25	73	83
New family	step 6	convex	M5+RM			12	–	100

Selection by the smallest decisive or safe distance among all rules or the pre-specified ones, or by the smallest decisive distance of step 6 (rule and band together); for a new family, selected on one transfer. Other: the distance on the other geometry. Correct, T: share correct among decided verdicts and trial rate there (%). Source: `qual_crossfit`.